



Boat Electrical Survey

Performed by Joe Zeitouny

Created on [REDACTED]

Boat's Name: [REDACTED]
Brand & Type: [REDACTED]
Build Year: [REDACTED]

Description

This report details a comprehensive electrical system survey conducted on the vessel [REDACTED], on [REDACTED]. The objective of this inspection was to assess the current health, functionality, and integrity of all onboard electrical components. This document covers visual inspections, operational testing, and diagnostic readings across key systems. Findings are categorized into Operational Status (Green/Yellow/Red), providing an immediate overview of system performance. Detailed sections include Battery Management Systems (BMS), Charging Infrastructure, Navigation & Communication Electronics, Lighting Circuits, and Auxiliary Power Distribution. Recommendations are provided to ensure optimal reliability, efficiency, and longevity of the vessel.



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1. Power Generation & Charging Systems (The "Input")

Topic	Status	Observations
Alternator / Generator Output - Measure output voltage under load (engine running). - Measure amperage output (load test). - Check for excessive heat or unusual noise during operation.		13.9 V when engine is running 30 A at output terminal - Temperature around 90 °C
Battery Bank Health - Measure resting voltage of each battery (individual and bank total). - Perform State of Charge (SoC) check (if possible via BMS/meter). - Check cell balance (for Lithium systems). - Inspect terminals for corrosion, tightness, and signs of overheating.		House bank AGM: - House Batt 1. 12.7 V 78% healthy - House Batt 2. 12.4 V 60% healthy Engine bank Gel: - Starter Batt. 12.6 V 70% healthy
Charging System Integrity - Verify charging voltage matches the battery type (e.g., 14.4V for LiFePO4). - Check charging current flow while running/at anchor. - Inspect shore power inlet connection and breaker integrity.		Shore power connection is corroded (see fig. 1 in the appendix below) and needs to be replaced
Inverter/Charger Performance - Test inverter output voltage stability (clean sine wave check if possible). - Verify automatic switching between battery, alternator, and shore power sources. - Check for proper float/maintenance charge settings.		Shore power charger improperly configured for charging the AGM type house battery bank



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2. Distribution & Wiring Integrity (The "Plumbing")

Topic	Status	Observations
Main Bus Bars & Fuses - Visually inspect main bus bars for signs of scorching or pitting. - Test fuses/breakers: Check continuity and trip mechanism functionality. - Check terminal tightness at all major connection points.	Yellow	Negative bus bar shows signs of corrosion needs to be cleaned and greased again (see fig 2. in the appendix below)
Cable Condition - Inspect visible runs for chafing, cracking, melting, or insulation damage (especially near engine beds/through-hulls). - Check cable routing: Ensure cables are not pinched, kinked, or resting on sharp edges.	Green	Cable runs seem to be ok although some minor chafing is observed on the cables right under the chart table
Grounding System - Measure resistance between the main negative bus and the hull/engine block (should be very low). - Verify all major components (navigation gear, engine electronics) have a dedicated, secure ground path.	Red	Resistance between negative bus bar and engine is 5 Ohms (OK) Rudder shaft needs to be grounded with the engine block All thruhull valves need to be grounded with the engine block Mast foot need to be grounded with the engine block



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3. Electronic Systems & Navigation (The "Brain")

Topic	Status	Observations
Navigation Electronics - GPS / Chartplotter: Verify signal lock and accuracy. Check display brightness/functionality. - Depth Sounder / Sonar: Test ping return strength and depth reading consistency. - Radar (if equipped): Check rotation speed, power draw, and screen clarity.		Reported depth needs to be offset to match the depth from the waterline to the sea floor
Communication Systems VHF Radio: Test transmit/receive functionality; check antenna connection integrity. AIS: Verify signal reception and display accuracy.		AIS unit is not transmitting location
Autopilot System - Test steering response under manual control vs. autopilot engagement. - Check sensor inputs (wind speed, heading) for drift or error.		Autopilot drive unit is broken Rudder feedback sensor is disconnected from the NMEA2000 network



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4. Control & Auxiliary Systems (The "Muscles")

Topic	Status	Observations
Lighting Circuits • Test all interior lights (cabin, galley, heads). • Test exterior navigation/safety lights (running lights, anchor light, courtesy lights). • Check for flickering or dimming under load.		Masthead tricolor light is not working
Engine Monitoring & Control • Verify engine gauge accuracy (RPM, Oil Pressure, Coolant Temp). • Check wiring harness integrity between the engine and control panel. • Test throttle/shift controls for smooth operation.		All looks ok
Pumps & Thrusters • Test bilge pumps: Check activation switch function and flow rate. • Test through-hull fittings: Ensure seals are intact and wiring is secure. • (If applicable) Test bow/stern thruster functionality and control response.		The bilge pump in the engine compartment is working when manually started however the automatic start feature is not working due to a broken float switch



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5. Specialized Systems (As Applicable)

Topic	Status	Observations
Lithium Battery Management System (BMS) • Verify BMS communication with the main display unit. • Check individual cell voltage readings against manufacturer specs. • Confirm temperature monitoring is active and reporting correctly.	N/A	Not applicable
Water Maker/Desalinator • Test operational cycle; check water output flow rate. • Verify system power draw under load.		Could not test while as the filters were not installed and the system isn't ready to be started.
Remote Controls & Sensors • Test remote switches (e.g., cabin lights, pump activation). • Check sensor calibration (e.g., trim sensors, fuel level probes).		Fuel level sensor is reporting inaccurate reading and needs to be replaced
Generator	N/A	Not applicable



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6. Documentation vs Current State

Topic	Status	Observations
System Mapping Verification - Compare the physical wiring layout against any existing system diagrams. - Note any undocumented or "hidden" circuits.		Electrical systems on board show no modifications from the initial factory designs
Documentation Check - Verify that all major components have legible model numbers and serial numbers recorded. - Confirm fuses/breakers are correctly sized for their respective loads.		All fuses are correctly sized as per the original factory documentation



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7. Final Thoughts

Overall, the vessel's electrical systems have been found to be in good working condition at the time of this survey. The installation demonstrates competent craftsmanship and adheres to recognized marine electrical standards throughout most components tested (except for the grounding system).

However, several items requiring attention have been identified above in this report. While none of these issues constitute critical safety hazards that would prohibit operation, they should be addressed by the prospective owner to ensure continued reliability and optimal system performance. These matters range from minor maintenance items to moderate upgrades that would benefit the long-term integrity of the electrical installation.

It is recommended that the purchaser budget for these repairs/refurbishments prior to or shortly after acquisition. Given the age and utilization typical of vessels in this class, some ongoing monitoring and periodic servicing of the electrical systems will be necessary to maintain safe operations.

The buyer may wish to retain a qualified marine electrician to review the items noted herein and provide cost estimates for any recommended remedial work before completing the transaction.

As with any survey, conditions may change between the date of inspection and purchase. Systems should be tested upon delivery of the vessel to confirm continued proper function.



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I. Appendix

Pictures taken from the vessel are documented in this section.